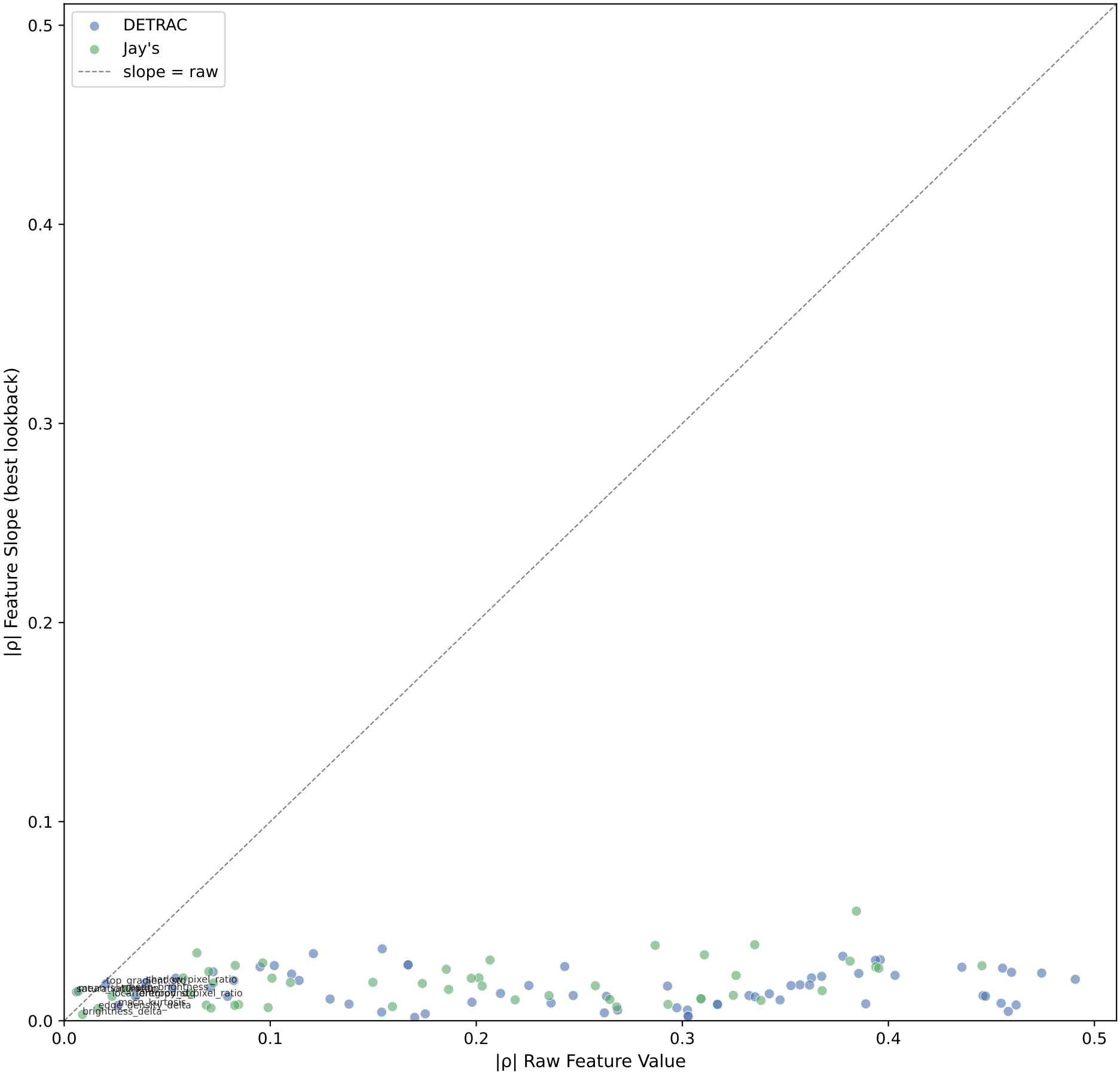
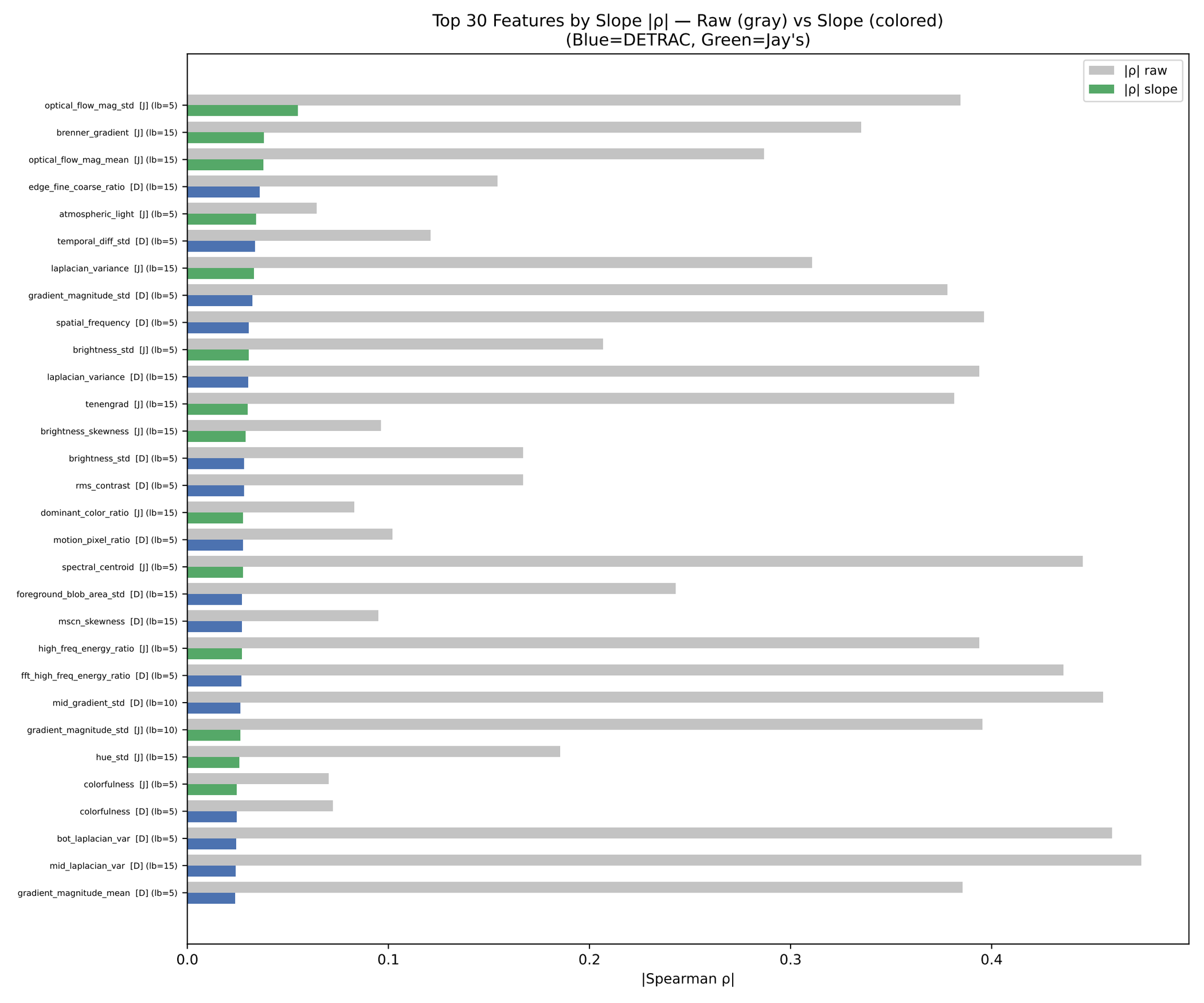
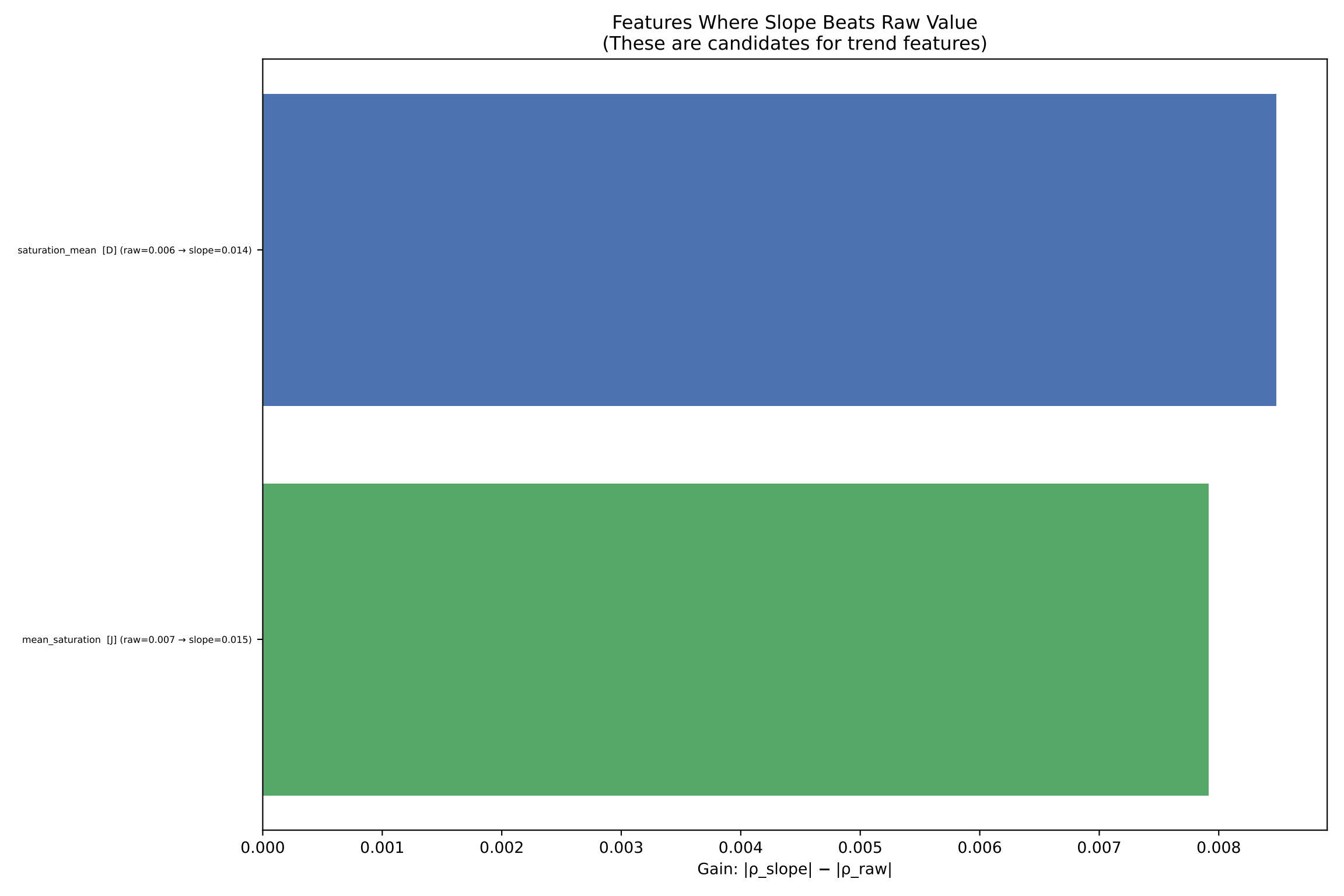
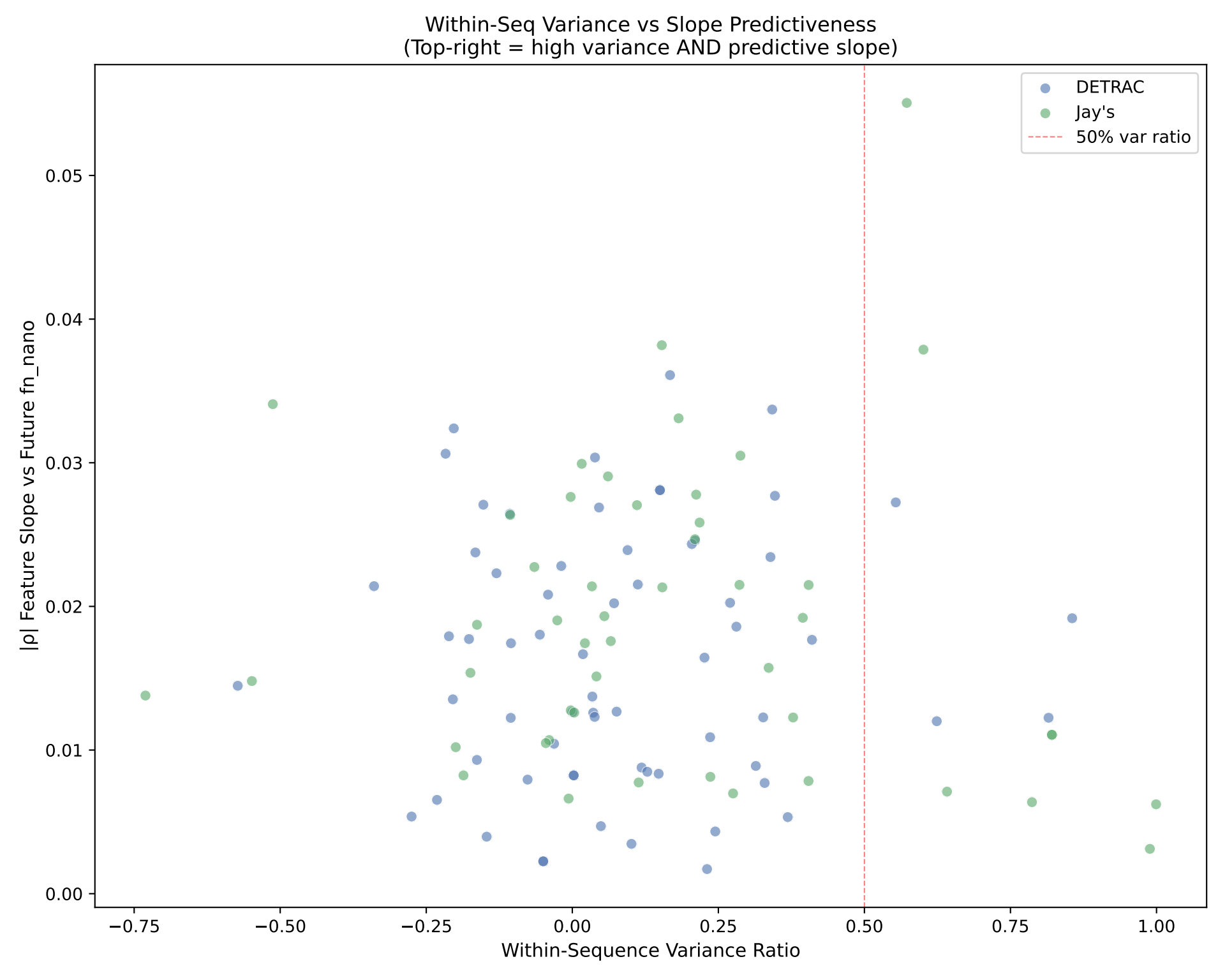
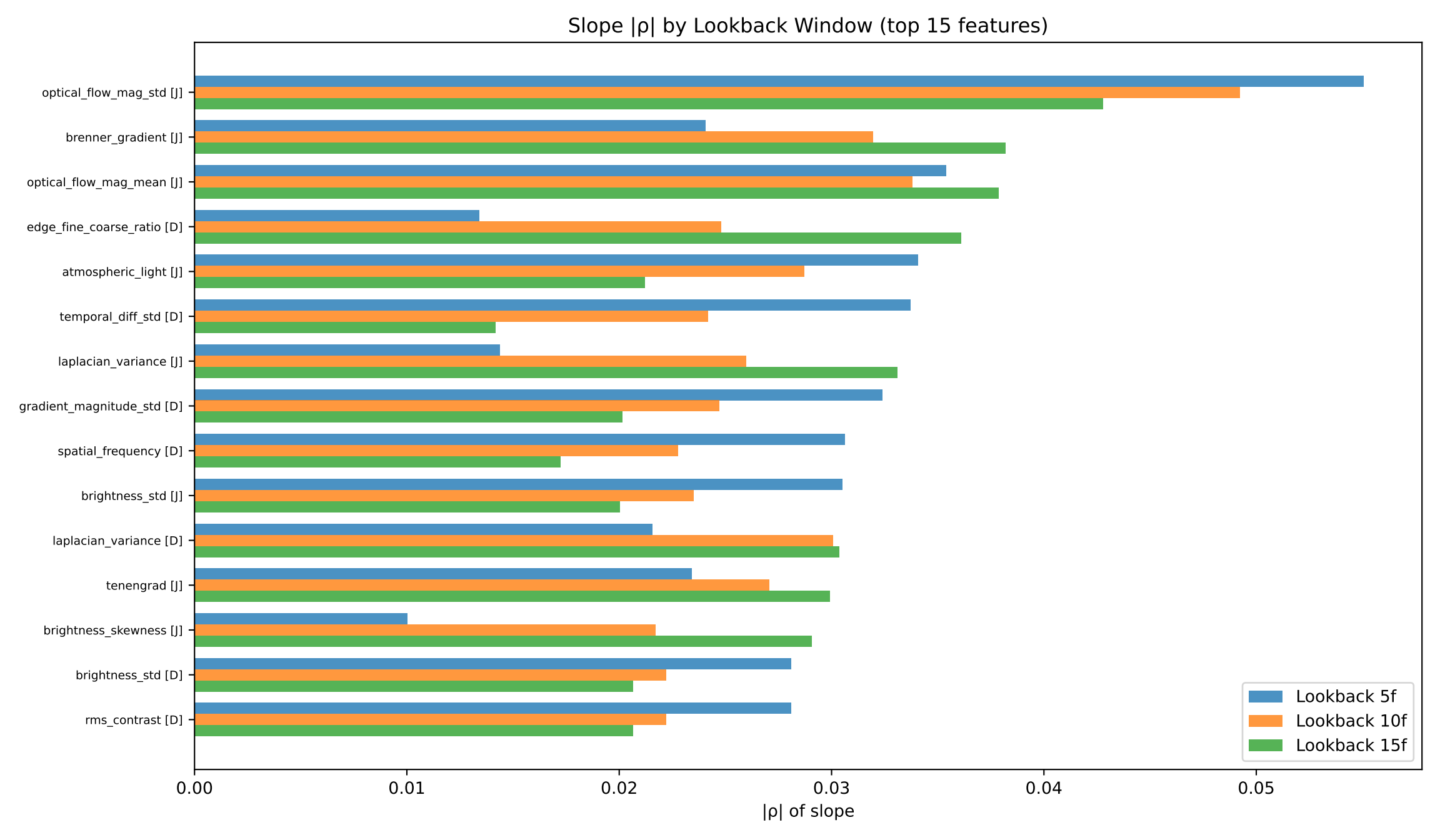

VisDrone Trend Analysis: Raw vs Slope Correlation with Future fn_nano
(Points above diagonal = slope beats raw)
DETRAC
Jay's
slope = raw
|ρ| Feature Slope (best lookback)
|ρ| Raw Feature Value
0.0
0.1
0.2
0.3
0.4
0.5
top_gradient_pixel_ratio
edge_density_delta
brightness_delta


Top 30 Features by Slope |ρ| — Raw (gray) vs Slope (colored)
(Blue=DETRAC, Green=Jay's)
|ρ| raw
|ρ| slope
optical_flow_mag_std [J] (lb=5)
brenner_gradient [J] (lb=15)
optical_flow_mag_mean [J] (lb=15)
edge_fine_coarse_ratio [D] (lb=15)
atmospheric_light [J] (lb=5)
temporal_diff_std [D] (lb=5)
laplacian_variance [J] (lb=15)
gradient_magnitude_std [D] (lb=5)
spatial_frequency [D] (lb=5)
brightness_std [J] (lb=5)
laplacian_variance [D] (lb=15)
tenengrad [J] (lb=15)
brightness_skewness [J] (lb=15)
brightness_std [D] (lb=5)
rms_contrast [D] (lb=5)
dominant_color_ratio [J] (lb=15)
motion_pixel_ratio [D] (lb=5)
spectral_centroid [J] (lb=5)
foreground_blob_area_std [D] (lb=15)
mscn_skewness [D] (lb=15)
high_freq_energy_ratio [J] (lb=5)
fft_high_freq_energy_ratio [D] (lb=5)
mid_gradient_std [D] (lb=10)
gradient_magnitude_std [J] (lb=10)
hue_std [J] (lb=15)
colorfulness [J] (lb=5)
colorfulness [D] (lb=5)
bot_laplacian_var [D] (lb=5)
mid_laplacian_var [D] (lb=15)
gradient_magnitude_mean [D] (lb=5)
0.0
0.1
0.2
0.3
0.4
|Spearman ρ|


Features Where Slope Beats Raw Value
(These are candidates for trend features)
saturation_mean [D] (raw=0.006 → slope=0.014)
mean_saturation [J] (raw=0.007 → slope=0.015)
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
Gain: |ρ_slope| − |ρ_raw|


Within-Seq Variance vs Slope Predictiveness
(Top-right = high variance AND predictive slope)
DETRAC
Jay's
50% var ratio
|ρ| Feature Slope vs Future fn_nano
0.00
0.01
0.02
0.03
0.04
0.05
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Within-Sequence Variance Ratio


Slope |ρ| by Lookback Window (top 15 features)
optical_flow_mag_std [J]
brenner_gradient [J]
optical_flow_mag_mean [J]
edge_fine_coarse_ratio [D]
atmospheric_light [J]
temporal_diff_std [D]
laplacian_variance [J]
gradient_magnitude_std [D]
spatial_frequency [D]
brightness_std [J]
laplacian_variance [D]
tenengrad [J]
brightness_skewness [J]
brightness_std [D]
rms_contrast [D]
Lookback 5f
Lookback 10f
Lookback 15f
0.00
0.01
0.02
0.03
0.04
0.05
|ρ| of slope

VisDrone Trend Analysis — Top 20 Features by Slope |ρ|
(Green rows = slope beats raw)

Feature	Source	ρ Raw	ρ Slope	Gain	Best LB	WS Var
optical_flow_mag_std	Jays	-0.3845	0.0551	-0.3294	5	0.573
brenner_gradient	Jays	+0.3351	0.0382	-0.2969	15	0.153
optical_flow_mag_mean	Jays	-0.2868	0.0379	-0.2489	15	0.601
edge_fine_coarse_ratio	DETRAC	-0.1544	0.0361	-0.1183	15	0.167
atmospheric_light	Jays	-0.0644	0.0341	-0.0303	5	-0.513
temporal_diff_std	DETRAC	-0.1210	0.0337	-0.0873	5	0.342
laplacian_variance	Jays	+0.3107	0.0331	-0.2776	15	0.182
gradient_magnitude_std	DETRAC	+0.3779	0.0324	-0.3455	5	-0.203
spatial_frequency	DETRAC	+0.3961	0.0306	-0.3655	5	-0.217
brightness_std	Jays	+0.2067	0.0305	-0.1762	5	0.288
laplacian_variance	DETRAC	+0.3937	0.0304	-0.3634	15	0.039
tenengrad	Jays	+0.3814	0.0299	-0.3515	15	0.016
brightness_skewness	Jays	-0.0964	0.0291	-0.0673	15	0.061
brightness_std	DETRAC	+0.1669	0.0281	-0.1388	5	0.150
rms_contrast	DETRAC	+0.1669	0.0281	-0.1388	5	0.150
dominant_color_ratio	Jays	-0.0830	0.0278	-0.0553	15	0.212
motion_pixel_ratio	DETRAC	-0.1020	0.0277	-0.0743	5	0.347
spectral_centroid	Jays	+0.4454	0.0276	-0.4178	5	-0.003
foreground_blob_area_std	DETRAC	-0.2429	0.0272	-0.2157	15	0.554
mscn_skewness	DETRAC	+0.0951	0.0271	-0.0680	15	-0.152